E1 244 Detection and Estimation Theory
Assignment 7

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Problem 1

Ans:

$$\mathbb{E}[C(H_0,\theta) \mid y = y] = \int_{\theta\in\Theta} C(H_0,\theta) f(\theta \mid y)\, d\theta$$

$$= \int_{\theta\in X_0} C(H_0,\theta) f(\theta \mid y)\, d\theta + \int_{\theta\in X_1} C(H_0,\theta) f(\theta \mid y)\, d\theta$$

$$= \int_{0.5}^{1} \frac{1}{\theta}\, f(\theta \in X_1 \mid y)\, d\theta.$$

$$f(\theta \in X_1 \mid y) = \frac{f(y \mid \theta \in X_1) f(\theta \in X_1)}{f(y)}$$

$$= \frac{f(y \mid \theta \in X_1, H_1)\, f(\theta \in X_1 \mid H_1)\, f(H_1)}{f(y)}.$$

So,

$$\mathbb{E}[C(H_0,\theta) \mid y = y] = \int_{0.5}^{1} \frac{1}{\theta} \cdot \theta e^{-\theta y} \mathbf{1}_{\{y\geq 0\}} \cdot 2\mathbf{1}_{\{0.5\leq\theta\leq 1\}} \cdot \frac{1}{2} \cdot \frac{1}{f(y)}\, d\theta$$

$$= \frac{\mathbf{1}_{\{y\geq 0\}}}{f(y)} \int_{0.5}^{1} e^{-\theta y}\, d\theta = \frac{\mathbf{1}_{\{y\geq 0\}}}{f(y)} \left[\frac{e^{-\theta y}}{-y}\right]_{0.5}^{1}$$

$$= \frac{\mathbf{1}_{\{y\geq 0\}}}{y f(y)} \left(e^{-y/2} - e^{-y}\right).$$

Also,

$$\mathbb{E}[C(H_1,\theta) \mid y = y] = \int_{\theta\in X_0} f(\theta \mid y)\, d\theta = \frac{f(y \mid \theta)}{f(y)} = \frac{1}{2}\frac{e^{-y/2}}{f(y)} \mathbf{1}_{\{y\geq 0\}}.$$

So, the Bayes rule:

$$\frac{e^{-y/2}(1-e^{-y/2})\mathbf{1}_{\{y\geq 0\}}}{yf(y)} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{1}{2}\frac{e^{-y/2}}{f(y)}\mathbf{1}_{\{y\geq 0\}}$$

$$\Rightarrow \quad \frac{1-e^{-y/2}}{y/2} \underset{H_0}{\overset{H_1}{\gtrless}} 1, \quad y \geq 0.$$

$$\frac{1-e^{-y/2}}{y/2} < 1, \ \forall y > 0, \qquad \text{at } y = 0 \text{ it is a tie.}$$

So,

$$\forall y > 0, \ \text{choose } \delta(y) = H_0; \qquad \text{at } y = 0, \ \text{choose } \delta(y) = H_1 \ \text{(any choice)}.$$

$$\bar{J}(y) = \begin{cases} \mathbb{E}[C(H_0,\theta) \mid y = y], & \delta(y) = H_0, \\ \mathbb{E}[C(H_1,\theta) \mid y = y], & \delta(y) = H_1. \end{cases}$$

$$J(\delta) = \int_{R_0} \bar{J}(y) f(y)\, dy + \int_{R_1} \bar{J}(y) f(y)\, dy,$$

where

$$R_0 = \{y : \delta(y) = H_0\}, \quad R_1 = \{y : \delta(y) = H_1\}.$$

Thus,

$$J(\delta) = \int_{R_0} \mathbb{E}[C(H_0,\theta) \mid y = y] f(y)\, dy + \int_{R_1} \mathbb{E}[C(H_1,\theta) \mid y = y] f(y)\, dy$$

$$= \int_0^\infty \frac{e^{-y/2} - e^{-y}}{yf(y)} f(y)\, dy = \int_0^\infty \left(\frac{e^{-y/2}}{y} - \frac{e^{-y}}{y} \right) dy.$$

Problem 2

Ans:

(a) $H_0 : \theta = 0$, $H_1 : \theta = \theta_1$.

$$f(y \mid H_0) = \prod_{i=1}^N f(y_i \mid H_0) = \frac{1}{(2\pi)^{N/2}} \exp\left(-\frac{1}{2} \sum_{i=1}^N y_i^2 \right).$$

$$f(y \mid H_1) = \prod_{i=1}^{N} f(y_i \mid H_1) = \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^{N} \frac{y_i^2}{1+\theta_1^2 s_i^2}\right)}{(2\pi)^{N/2}\prod_{i=1}^{N}(1+\theta_1^2 s_i^2)^{1/2}}.$$

$$L(y) = \frac{f(y \mid H_1)}{f(y \mid H_0)} = \frac{\exp\left(-\frac{1}{2}\left[\sum_{i=1}^{N} \frac{y_i^2}{1+\theta_1^2 s_i^2} - \sum_{i=1}^{N} y_i^2\right]\right)}{\prod_{i=1}^{N}(1+\theta_1^2 s_i^2)^{1/2}}.$$

$$L(y) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

$$\Rightarrow \quad -\frac{1}{2}\sum_{i=1}^{N}\left[\frac{y_i^2}{1+\theta_1^2 s_i^2} - y_i^2\right] - \frac{1}{2}\sum_{i=1}^{N}\ln(1+\theta_1^2 s_i^2) \underset{H_0}{\overset{H_1}{\gtrless}} \ln\gamma$$

$$\Rightarrow \quad \sum_{i=1}^{N} y_i^2\left[1 - \frac{1}{1+\theta_1^2 s_i^2}\right] \underset{H_0}{\overset{H_1}{\gtrless}} 2\ln\gamma + \sum_{i=1}^{N}\ln(1+\theta_1^2 s_i^2).$$

Define

$$T(y) = \sum_{i=1}^{N} \frac{\theta_1^2 s_i^2}{1+\theta_1^2 s_i^2} y_i^2, \qquad \eta = 2\ln\gamma + \sum_{i=1}^{N}\ln(1+\theta_1^2 s_i^2).$$

So NP test is $T(y) \underset{H_0}{\overset{H_1}{\gtrless}} \eta$.

$$T(y) = \sum_{i=1}^{N} a_i y_i^2, \quad a_i = \frac{\theta_1^2 s_i^2}{1+\theta_1^2 s_i^2},$$

which is generalized chi-square.

Let

$$z_i = \frac{y_i}{\sqrt{1+\theta_1^2 s_i^2}}, \quad z_i \sim \mathcal{N}(0,1),$$

then

$$T(y) = \sum_{i=1}^{N} \theta_1^2 s_i^2 z_i^2 = \sum_{i=1}^{N} b_i z_i^2, \quad b_i = \theta_1^2 s_i^2.$$

$$P_F = \int_{\eta}^{\infty} f_{T|H_0}(t)\,dt, \qquad P_D(\theta) = \int_{\eta}^{\infty} f_{T|H_1}(t)\,dt.$$

So NP test: max P_D s.t. $P_F \le \alpha$.

(b) $H_0 : \theta = 0$ versus $H_1 : \theta > 0$.

We have scalar parameter θ. For any $\theta_0 \in \bar{\Theta}_0 = \{0\}$, $\theta_1 \in \bar{\Theta}_1 = [0, \infty)$,

$$L(y; \theta_0, \theta_1) = \frac{f(y \mid \theta_1)}{f(y \mid \theta_0)} = \frac{\exp\left(\frac{1}{2}\sum_{i=1}^N y_i^2 \frac{\theta_1^2 s_i^2}{1+\theta_1^2 s_i^2}\right)}{\prod_{i=1}^N (1+\theta_1^2 s_i^2)^{1/2}} \quad \text{(using } \theta_1 \text{ constant).}$$

The UMP would exist when $s_1 = s_2 = \cdots = s_N = s > 0$. Then $T(y) = \sum y_i^2$ and $L(y; \theta_0, \theta_1)$ is a non-decreasing function in $T(y)$.

Then UMP exists of the form

$$\sum_{i=1}^N y_i^2 \underset{H_0}{\overset{H_1}{\gtrless}} \gamma,$$

where γ is selected s.t. $P_F = \alpha$:

$$P_F = \int_\gamma^\infty f(T \mid H_0)\, dT = 1 - \text{cdf of chi-square}(\gamma).$$

So,

$$\gamma = F_N^{-1}(1 - P_F) \quad [F_N \text{ is cdf of chi-square}],$$

and

$$T(y) \underset{H_0}{\overset{H_1}{\gtrless}} F_N^{-1}(1-\alpha).$$

(c) GLRT assuming $s_1 = s_2 = \cdots = s_N = s$.

$$\hat{\theta}_0 = 0, \qquad \hat{\theta}_1 = \arg\sup_{\theta \in \Theta_1} f(y \mid \theta) = \arg\sup_{\theta \in \Theta_1} \frac{\exp\left(-\frac{1}{2(1+\theta^2 s^2)}\sum_{i=1}^N y_i^2\right)}{(2\pi)^{N/2}(1+\theta^2 s^2)^{N/2}}.$$

Differentiating and setting to 0, with

$$p^2 = \frac{1}{1+\theta^2 s^2}, \qquad T(y) = \sum y_i^2,$$

we get (as in derivation)

$$N = T(y)\, p \Rightarrow p = \frac{N}{T(y)} \Rightarrow 1 + \theta^2 s^2 = \frac{T(y)}{N} \Rightarrow \theta = \frac{1}{s^2}\left(1 - \frac{T(y)}{N}\right).$$

GLRT:

$$\frac{f(y \mid \hat{\theta}_1)}{f(y \mid \hat{\theta}_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

$$\frac{\left(\frac{N}{T(y)}\right)^{N/2} \exp\left(-\frac{1}{2}\frac{N}{T(y)}\,T(y)\right)}{\exp\left(-\frac{1}{2}T(y)\right)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

$$\left(\frac{N}{T(y)}\right)^{N/2} \exp\left(-\frac{1}{2}N\right) \exp\left(\frac{1}{2}T(y)\right) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

$$\frac{e^{-\frac{1}{2}T(y)}}{T(y)^{N/2}} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{\gamma}{\exp\left(-\frac{1}{2}N\right) N^{N/2}} \triangleq \eta.$$

For $T \geq 0$, $\dfrac{e^{-T/2}}{T^{N/2}}$ is a decreasing function. So final test can be written as

$$T \underset{H_1}{\overset{H_0}{\gtrless}} T^*,$$

where T^* is such that

$$e^{-\frac{1}{2}T^*} = \eta\,(T^*)^{N/2}.$$

Problem 3

Ans:

(a)

$$y = H\theta + w.$$

When $\theta \neq 0$, $y \mid \theta \sim \mathcal{N}(H\theta, \sigma^2 I)$. So,

$$f(y \mid \theta) = L(\theta) = k \exp\left[-\frac{1}{2}(y - H\theta)^T(\sigma^2 I)^{-1}(y - H\theta)\right].$$

$$\hat{\theta}_1(y) = \arg\max_{\theta\neq 0} f(y \mid \theta) = \arg\min_{\theta\neq 0} \|y - H\theta\|_2^2 = (H^T H)^{-1} H^T y.$$

(b) GLRT

$$\frac{\sup_{\theta\neq 0} f(y \mid \theta)}{f(y \mid 0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma, \qquad f(y \mid 0) = k\, e^{-\frac{1}{2\sigma^2}\|y\|_2^2}.$$

$$\sup_{\theta\neq 0} f(y \mid \theta) = k\, e^{-\frac{1}{2\sigma^2}\|y - H\hat{\theta}(y)\|_2^2}.$$

So,

$$e^{-\frac{1}{2\sigma^2}\|y-H\hat{\theta}(y)\|_2^2+\frac{1}{2\sigma^2}\|y\|_2^2} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

$$\Rightarrow \quad \frac{1}{2\sigma^2}\left(y^Ty - y^Ty + 2\hat{\theta}(y)^TH^Ty - \hat{\theta}(y)^TH^TH\hat{\theta}(y)\right) \underset{H_0}{\overset{H_1}{\gtrless}} \ln\gamma.$$

Since

$$2\hat{\theta}_1(y)^TH^Ty = \hat{\theta}_1(y)^TH^TH\hat{\theta}_1(y) \quad [\because\ y - H\hat{\theta}\ \perp\ \mathrm{col}(H)],$$

we get

$$s(y) = \frac{\hat{\theta}_1^T(y)H^TH\hat{\theta}_1(y)}{\sigma^2} \underset{H_0}{\overset{H_1}{\gtrless}} \eta.$$

(c)

$$\hat{\theta}(y) = (H^TH)^{-1}H^Ty.$$

If $y \sim \mathcal{N}(\mu, \sigma^2 I)$, then for any A:

$$Ay \sim \mathcal{N}(A\mu, \sigma^2 AA^T).$$

Hence

$$\hat{\theta}(y) \sim \mathcal{N}\left((H^TH)^{-1}H^T\mu,\ (H^TH)^{-1}\sigma^2\right).$$

When $\theta = 0$,

$$\hat{\theta}(y) \sim \mathcal{N}\left(0, (H^TH)^{-1}\sigma^2\right).$$

By property given in hint,

$$\frac{\hat{\theta}^T(y)(H^TH)\hat{\theta}(y)}{\sigma^2} \text{ is chi-squared with } M \text{ degrees of freedom.}$$

So,

$$s(y) \sim \chi^2_M \text{ under } H_0.$$

When $\mu \neq 0$, similarly $s(y)$ is non-central chi-squared, with noncentrality

$$\left((H^TH)^{-1}H^TH\theta\right)^T\left(\frac{H^TH}{\sigma^2}\right)\left((H^TH)^{-1}H^TH\theta\right) = \frac{\theta^TH^TH\theta}{\sigma^2}.$$

Now, if R is P.D., by Cholesky $R = LL^T$. Define $z = L^{-1}x$. Then

$$z \sim \mathcal{N}(L^{-1}\mu, I).$$

Also,

$$x^T R^{-1} x = x^T (L^T)^{-1} L^{-1} x = z^T z = \sum z_i^2.$$

If $\mu = 0$, then $x^T R^{-1} x$ is a χ^2_M distribution. If $\mu \neq 0$, then $x^T R^{-1} x$ is a non-central χ^2_M distribution with

$$\lambda = \sum \mu_i'^2 = \mu'^T \mu', \quad \mu' = L^{-1}\mu.$$

Then,

$$\mu'^T (L^{-1})^T L^{-1} \mu = \mu^T R^{-1} \mu \Rightarrow \lambda = \mu^T R^{-1} \mu.$$

Problem 4

Ans:

(a)

<u>Test 1:</u>

$$P_F = \Pr(H_1 \mid H_0).$$

If $H_0 : \theta = 0$, then $X_1, X_2 \sim$ i.i.d. Uniform$[0, 1]$.

$$\int \mathbf{1}_{\{0.95 \leq x \leq 1\}} \, dx_1 = 0.05.$$

So, P_F for Test 1 is 0.05.

<u>Test 2:</u>

$$\Pr(X_1 + X_2 \geq c), \quad X_1, X_2 \sim \text{Uniform}(0, 1).$$

For $1 \leq c \leq 2$, the area is

$$\frac{1}{2}(2-c)(2-c) = \frac{(2-c)^2}{2}.$$

By problem,

$$\frac{(2-c)^2}{2} = \alpha \Rightarrow (2-c)^2 = 2\alpha \Rightarrow 2 - c = \pm\sqrt{2\alpha} \Rightarrow c = 2 \pm \sqrt{2\alpha}.$$

Since $c \leq 2$, we take

$$c = 2 - \sqrt{2\alpha}.$$

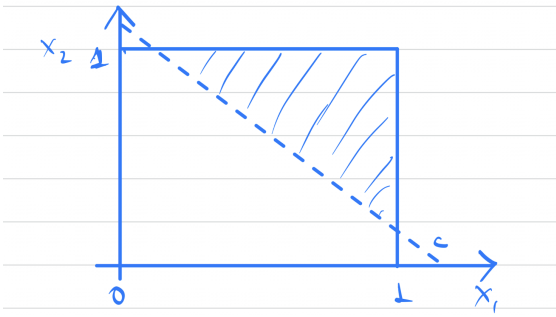

Figure 1: Area sketch used for computing $\Pr(X_1 + X_2 \geq c)$ in Problem 4(a).

(b)

<u>Test 1:</u>

$$P_D = \Pr(X_1 \geq 0.95 \mid H_1) = \theta + 1 - \max\{\theta, 0.95\} = \begin{cases} 1, & \theta \geq 0.95, \\ \theta + 0.05, & 0 \leq \theta < 0.95. \end{cases}$$

<u>Test 2:</u>

$$P_D = \Pr(X_1 + X_2 \geq c \mid H_1).$$

If $c \leq 2\theta$, $P_D(\theta) = 1$. If $c > 2\theta + 2$, $P_D(\theta) = 0$.
For intermediate ranges:

$$2\theta < c \leq 2\theta + 1 \quad \Rightarrow \quad P_D(\theta) = 1 - \frac{1}{2}(c - 2\theta)^2,$$

$$2\theta + 1 < c \leq 2\theta + 2 \quad \Rightarrow \quad P_D(\theta) = \frac{1}{2}(2\theta + 2 - c)^2.$$

So,

$$P_D(\theta) = \begin{cases} 1, & \theta \geq \frac{c}{2}, \\ 1 - \frac{1}{2}(c - 2\theta)^2, & \frac{c-1}{2} \leq \theta < \frac{c}{2}, \\ \frac{1}{2}(2\theta + 2 - c)^2, & \frac{c-2}{2} \leq \theta < \frac{c-1}{2}, \\ 0, & \theta < \frac{c-2}{2}. \end{cases}$$

(c)

Using

$$c = 2 - \sqrt{2\alpha} \approx 1.684.$$

Test 1 is not uniformly most powerful than Test 2.

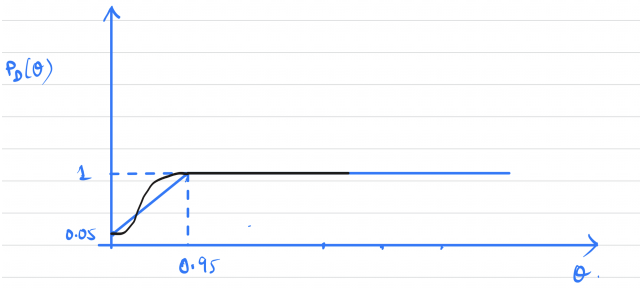

Figure 2: $P_D(\theta)$ comparison sketch for Test 1 and Test 2 (Problem 4(c)).